

Stat 301 Notes

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Course Goals for STAT 301

STAT 301 is an introduction to the theory underlying statistical inference. By the end of the course, you will:

- Understand different methods of obtaining point estimates, including maximum likelihood, method of moments, and Bayesian estimation
- Understand that some estimators are better than others, and be able to evaluate estimators
- Understand the theory underlying hypothesis tests, including errors, power, size, and level of tests
- Be able to derive hypothesis tests for simple situations from first principles, including likelihood ratio tests and most powerful tests
- Be able to derive interval estimates by inverting tests and using pivotal quantities

1 Order Statistics

During World War II, the Allied Forces analyzed the markings/serial numbers found on captured German equipment to see if they could determine the strength of the German forces. Specifically, they examined the serial numbers found on tanks. Allied spies were instructed to record the serial numbers of any German tanks they observed in the field. The Allied statisticians made an assumption: that the Germans had sequentially numbered their tanks in the order in which they were produced (this turned out to be true).

Without loss of generality, assume the serial numbers were $1, 2, 3, \dots, N$. When a tank was observed, the Allied forces recorded the serial number and developed a procedure to estimate N . Let's suppose that when one spy returned from her mission and was de-briefed, she reported the following observed serial numbers: 5, 12, 32, 43, 49, 54.

In small groups, discuss the following:

- What is the research question we want to answer?
- What is the parameter we want to estimate?
- What statistic can we use to estimate this parameter? What is your reasoning?

We're going to need to remember a **lot** of concepts from STAT 262. We'll start with this one:

💡 Multinomial Distribution

The multinomial distribution is characterized by:

1. n identical, independent trials
2. Each trial results in one of k outcomes
3. An outcome of type i has probability p_i , where $p_1 + p_2 + \dots + p_k = 1$ and p_i is constant from trial to trial
4. The random variables of interest are $Y_1, Y_2, \dots, Y_k$, where Y_i is the number of trials resulting in outcome i (Note: $Y_1 + \dots + Y_k = n$)

pmf:

$$f_{Y_1, \dots, Y_k}(y_1, \dots, y_k; n, p_1, \dots, p_k) = \begin{cases} \frac{n!}{y_1! y_2! \dots y_k!} p_1^{y_1} p_2^{y_2} \dots p_k^{y_k} & \begin{array}{l} y_i = 0, 1, 2, \dots, n \\ y_1 + \dots + y_k = n \\ i = 1, \dots, k \end{array} \\ 0 & \text{otherwise} \end{cases}$$

Why are we thinking about the multinomial? It turns out to be useful in finding the distribution of order statistics.

1.1 What are Order Statistics?

While the sample mean and sample variance can be very useful quantities in many situations, they are not always the most appropriate statistics for a particular research question. In the German tank problem, we were interested in estimating the total number of tanks the Germans had produced. A spy observed the following serial numbers: 5, 12, 32, 43, 49, 54.

- Why would the sample mean or sample variance be inappropriate here?
- Is the sample random? Why or why not?

If we want to make inferences about a parameter, we need to know the distribution of the statistic/estimator we're using. Last semester, we derived the distributions of the sample mean and sample variance, as well as the distributions of some functions of these statistics. Now, we'll turn our attention to the distributions of **order statistics**, such as the sample maximum. You've already seen these! You worked with them on the STAT 262 final project.

 Definition: Order statistics

The **order statistics** of a random sample $X_1, X_2, \dots, X_n$ are the sample values placed in ascending order. They are denoted by:

Example: A snow storm goes through Lincoln, and drops 10 inches of snow. Consider $n = 5$ randomly selected neighborhoods in Lincoln, and let X_i be the time (in hours, from the time the snow stops falling) until neighborhood i is visited by the city snowplows. Assuming independence (realistic?), the observed times are $(x_1, x_2, x_3, x_4, x_5) = (12, 7, 72, 33, 20)$. In this case, the actual observations for each neighborhood are denoted $x_2 = 7$, $x_1 = 12$, $x_5 = 20$, $x_4 = 33$, $x_3 = 72$. However, we can also consider the “ordered” random sample:

$$\begin{aligned} \text{In general, } X_{(1)} &= \\ X_{(2)} &= \\ &\vdots \\ X_{(n)} &= \end{aligned}$$

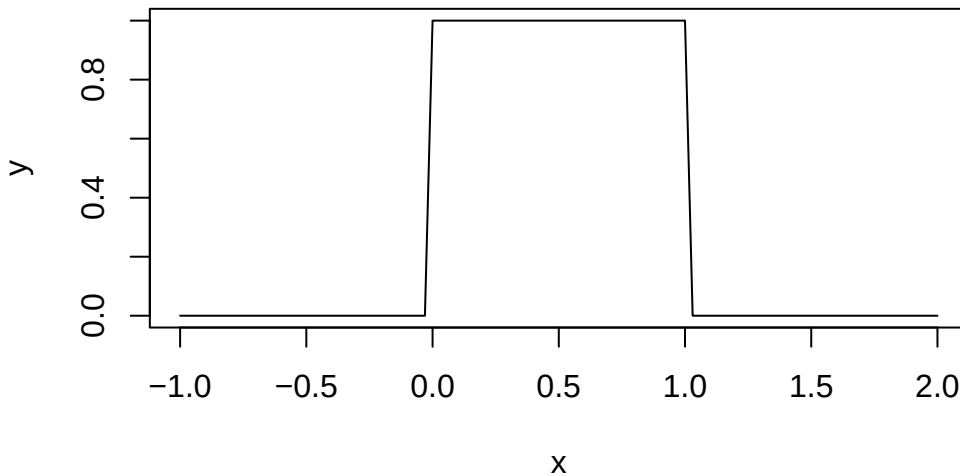
Some commonly used order statistics (and functions of order statistics) include:

However, in order to make inferences about a parameter, we need to know the distribution of the corresponding statistic/estimator. Why?

Let's consider some different distributions. Let's start with the Uniform.

1. Let $X_1, \dots, X_n$ be a random sample from the Uniform(0,1) distribution.

```
x<-seq(-1,2,length=100)
y<-dunif(x,min=0,max=1,)
plot(x,y,type='l')
```



Let's first consider $X_{(1)}$, from samples of both $n = 2$ and $n = 100$.

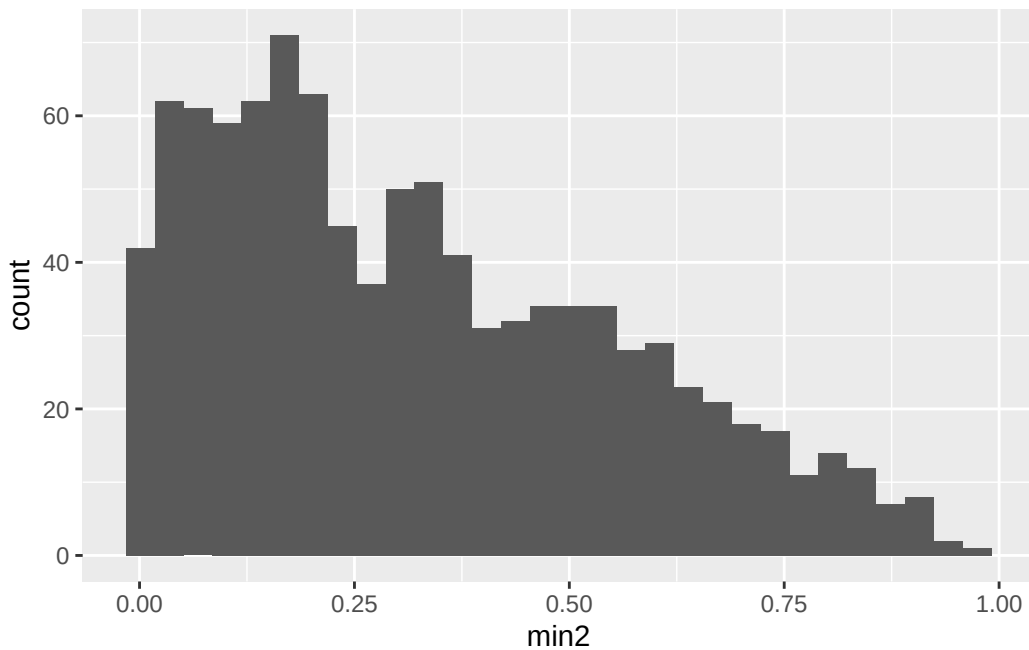
```
library(ggplot2)

unisample<-data.frame(sim=1:1000,min2=NA,min100=NA)

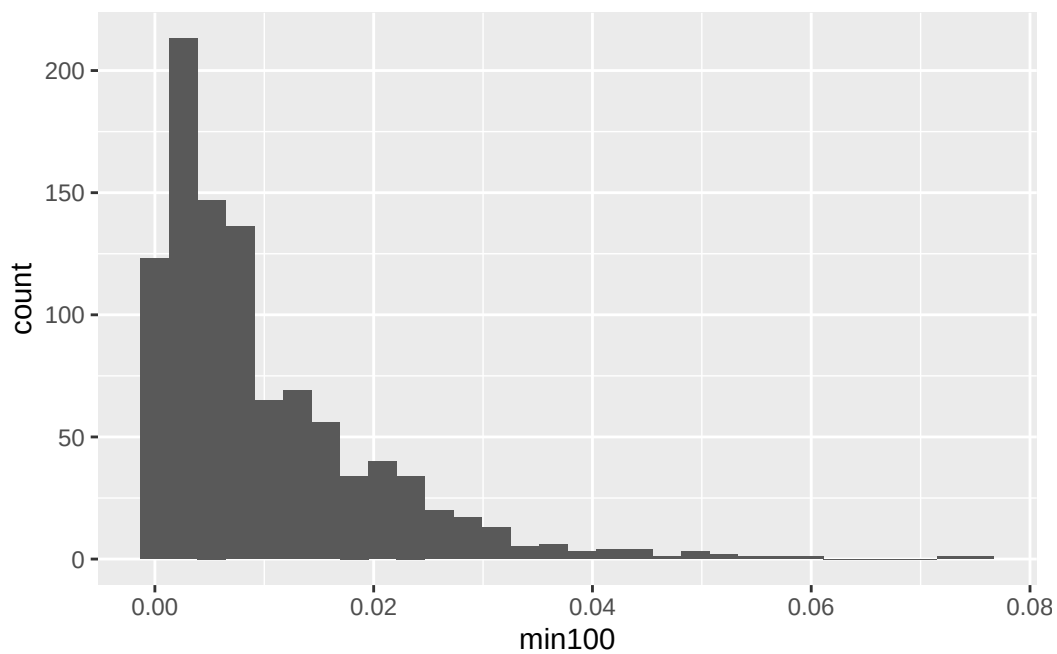
for(i in 1:1000){
  data1<-runif(2,min=0,max=1)
```

```
data2<-runif(100,min=0,max=1)
unisample$min2[i]<-min(data1)
unisample$min100[i]<-min(data2)
}
```

```
ggplot(unisample, aes(min2)) + geom_histogram()
```



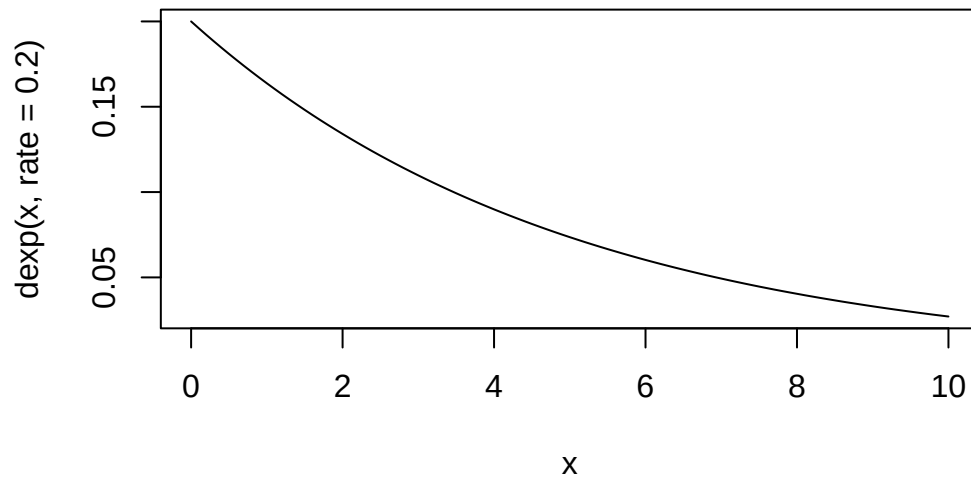
```
ggplot(unisample, aes(min100)) + geom_histogram()
```

What do you think the (empirical) distribution of the max, $X_{(n)}$ would look like?

2. Let $X_1, \dots, X_n$ be a random sample from the Exponential($\lambda = 5$) distribution.

```
curve(dexp(x,rate=0.2),from=0, to=10)
```



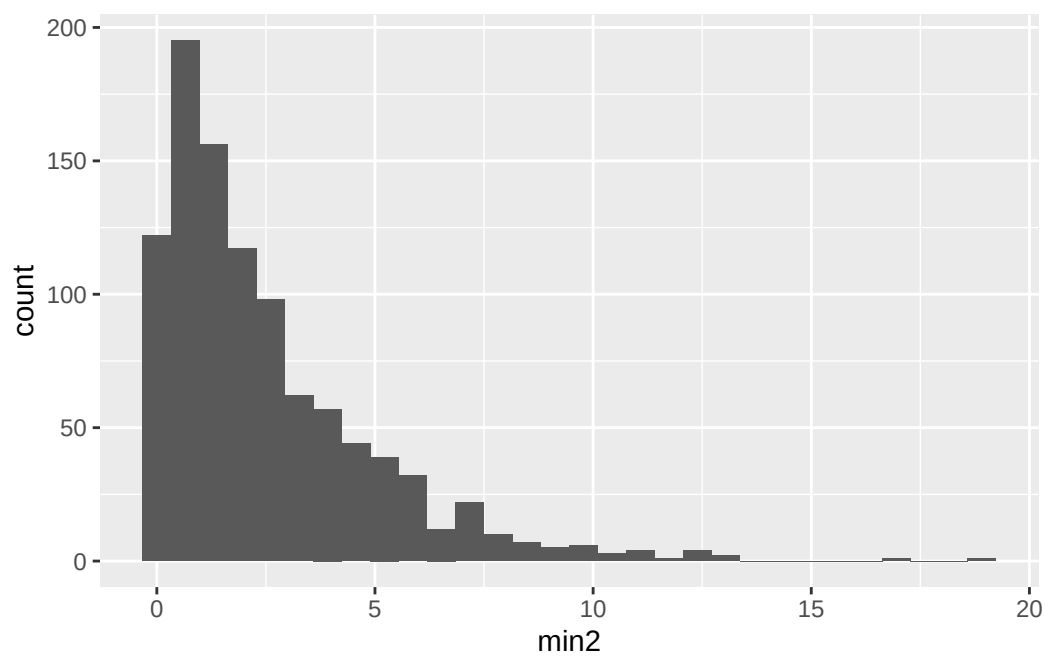
Again, let's consider $X_{(1)}$, from samples of both $n = 2$ and $n = 100$.

```
library(ggplot2)

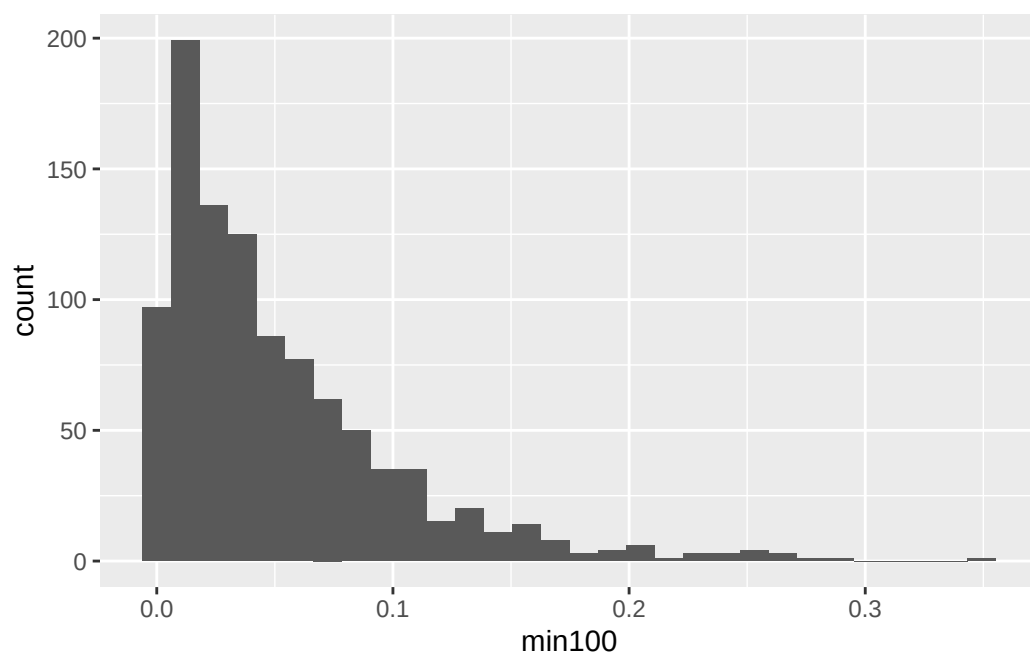
expsample<-data.frame(sim=1:1000,min2=NA,min100=NA)

for(i in 1:1000){
  data1<-rexp(2,rate=0.2)
  data2<-rexp(100,rate=0.2)
  expsample$min2[i]<-min(data1)
  expsample$min100[i]<-min(data2)
}

ggplot(expsample, aes(min2)) + geom_histogram()
```



```
ggplot(expsample, aes(min100)) + geom_histogram()
```



What about $X_{(n)}$ in this case?

What factors seems to influence what the distribution of an order statistic looks like?

1.2 The Distribution of an Order Statistic

We'll derive the theoretical distribution of a min and max through our uniform distribution example, and then we'll generalize to any distribution (and any order statistic).

Example: Let $X_1, \dots, X_n$ be a random sample from the Uniform(0,1) distribution.

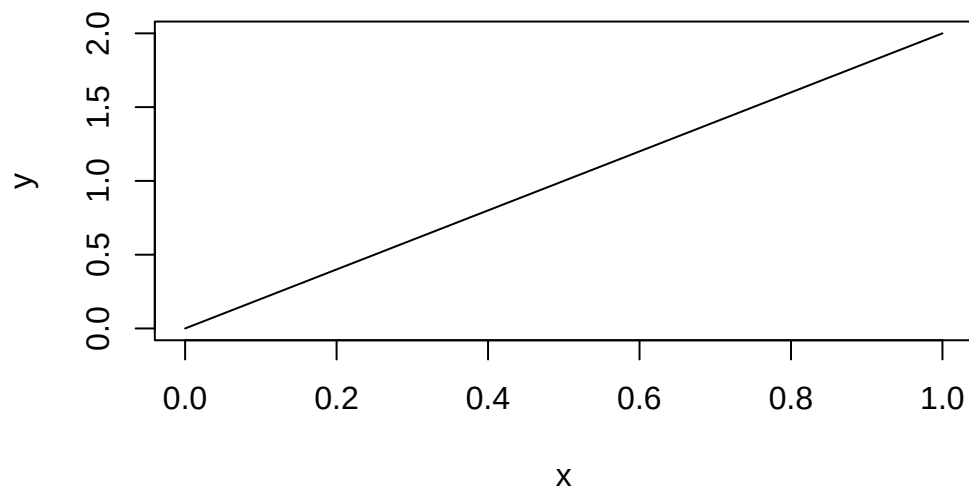
- What are the pdf and CDF of the parent population?

- Find the CDF of $X_{(n)}$

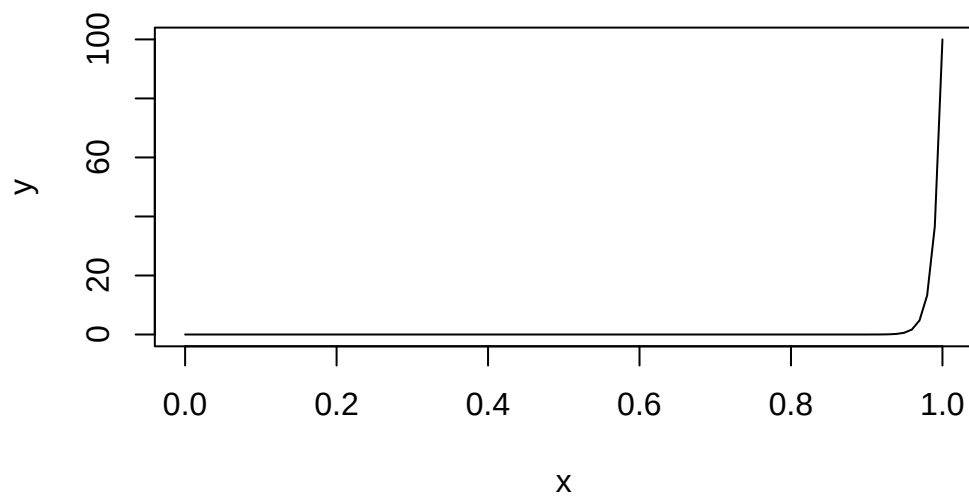
- Find the pdf of $X_{(n)}$

So, for a random sample of Uniform(0,1) random variables

- When $n = 2$, the pdf of $X_{(n)}$ is



- When $n = 100$, the pdf of $X_{(n)}$ is



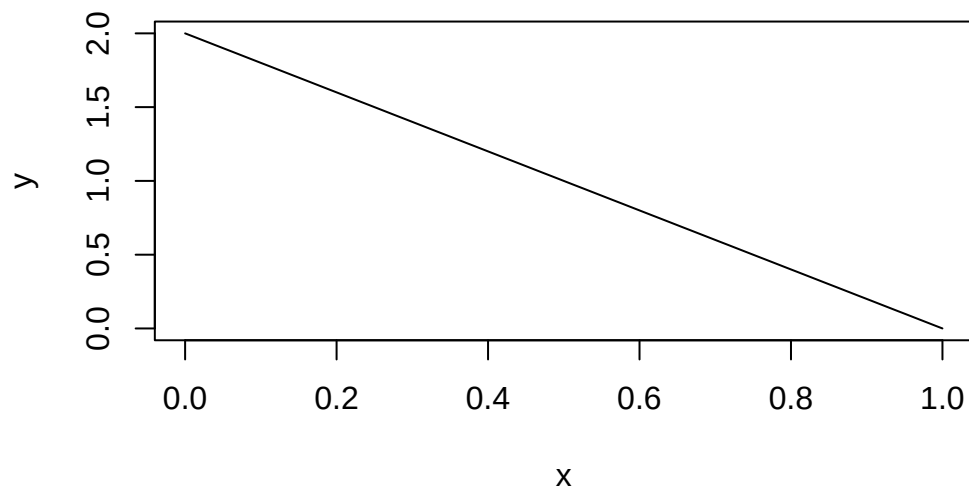
Example: Now, let's try the min.

- Find the CDF of $X_{(1)}$

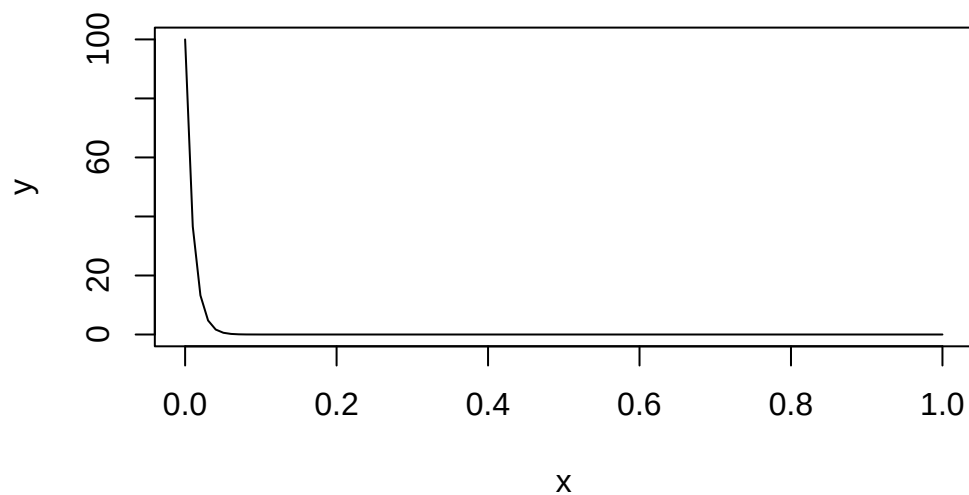
- Find the pdf of $X_{(1)}$

So, for a random sample of Uniform(0,1) random variables

- When $n = 2$, the pdf of $X_{(1)}$ is



- When $n = 100$, the pdf of $X_{(1)}$ is



What about the distributions of other order statistics, like the median? Is there a way we can cut to the pdf straightaway?

💡 Theorem: pdf of the j^{th} order statistic

Let $X_1, \dots, X_n$ be independent and identically distributed *continuous* random variables with CDF $F(x)$ and pdf $f(x)$. The pdf of $X_{(j)}$ is given by

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!} [F(x)]^{j-1} f(x) [1 - F(x)]^{n-j}$$

“Proof” (Please note that this is not a **real** proof) (and why haven’t we used the multinomial distribution, yet?) (and are we going to have to *memorize* this???)

- For the j^{th} order statistic, $X_{(j)}$, we have three “buckets” can toss each X_i into:
 - Group 1:
 - Group 2:
 - Group 3:

- What are the number of ways to select groups of size $j - 1$, 1, and $n - j$ from a sample of size n ?

- How “likely” is one of these ways?

- Putting these together:

Let's try out this theorem with the exponential distribution.

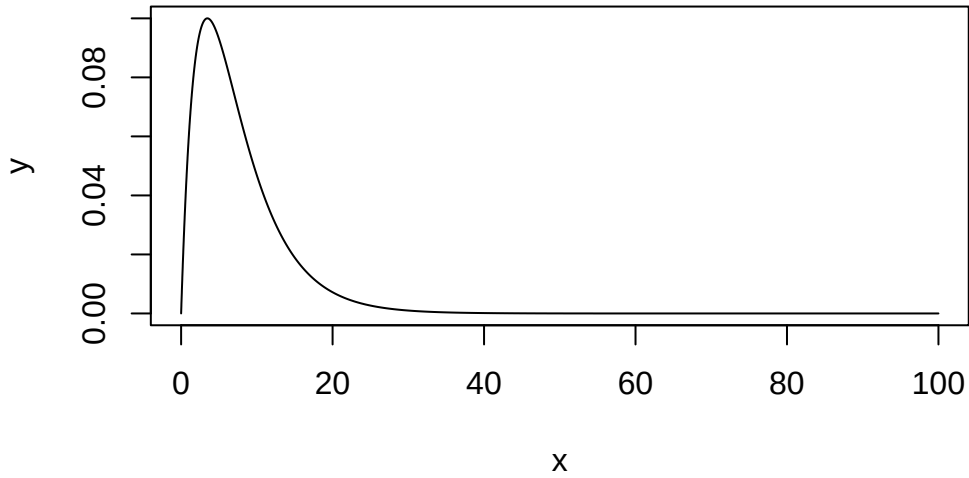
Example: Let $X_1, \dots, X_n$ be a random sample from the exponential($\lambda = 5$) distribution.

- What are the pdf and CDF of the parent population?

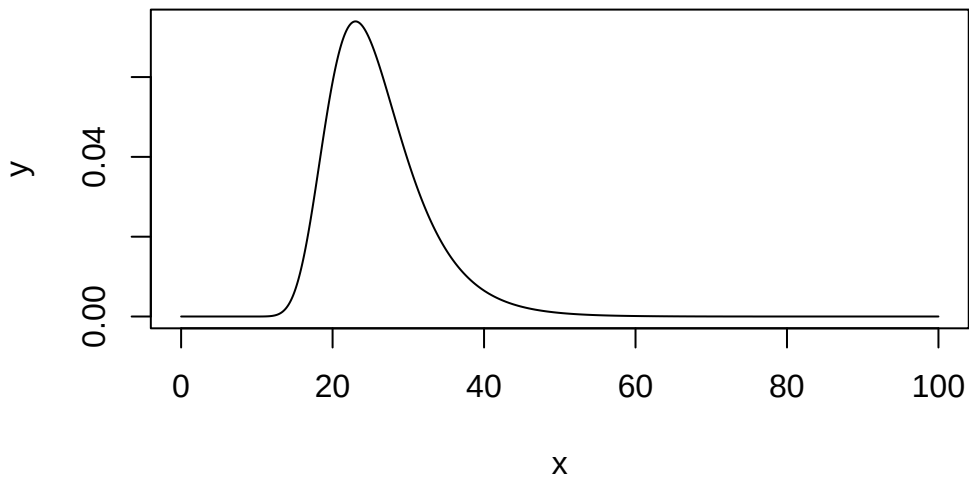
- Using the theorem, find the pdf of $X_{(n)}$

So, for a random sample of Exponential(5) random variables

- When $n = 2$, the pdf of $X_{(n)}$ is



- When $n = 100$, the pdf of $X_{(n)}$ is



We could also use this logic to find the joint distribution of any two order statistics or the joint distribution of all n order statistics.